

VERIFICATION OF WSL INSTALLATION:

```
Windows PowerShell
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PS C:\Users\aleky> wsl --status
Default Distribution: Ubuntu
Default Version: 2
WSL1 is not supported with your current machine configuration.
Please enable the "Windows Subsystem for Linux" optional component to use WSL1.
PS C:\Users\aleky>
```

VERIFICATION OF GCC VERSION:

```
alekhya@AlekhyaReddy: ~$ gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
Copyright (C) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

alekhya@AlekhyaReddy: ~$ |
```

GCC PROGRAM:

```
GNU nano 7.2 fork_exec.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%99s", command);

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed!\n");
        return 1;
    }
    else if (pid == 0)
    {
        printf("\n=== Child Process ===\n");
        printf("Child PID = %d\n", getpid());

        execlp(command, command, NULL);

        printf("Command execution failed!\n");
        exit(1);
    }
    else
    {
        printf("\n=== Parent Process ===\n");
        printf("Parent PID = %d\n", getpid());
        printf("Child PID = %d\n", pid);

        wait(NULL);

        printf("\nChild process completed.\n");
    }

    return 0;
}
```

```
alekhya@Alekhyareddy: ~  
alekhya@Alekhyareddy:~$ nano process.c  
alekhya@Alekhyareddy:~$ gcc process.c -o process  
alekhya@Alekhyareddy:~$ ./process  
Enter a Linux command: pwd  
  
Parent Process  
Parent PID : 8761  
Child PID : 8770  
  
Child Process  
Child PID : 8770  
Parent PID : 8761  
  
Executing command: pwd  
  
/home/alekhya  
  
Child process completed.  
alekhya@Alekhyareddy:~$
```

SIMPLE COMMANDS:

```
alekhya@Alekhyareddy: ~  
alekhya@Alekhyareddy:~$ pwd  
/home/alekhya  
alekhya@Alekhyareddy:~$ date  
Mon Aug 3 15:05:10 UTC 2026  
alekhya@Alekhyareddy:~$ whoami  
alekhya  
alekhya@Alekhyareddy:~$ hostname  
Alekhyaareddy  
alekhya@Alekhyareddy:~$ uname  
Linux  
alekhya@Alekhyareddy:~$ id  
uid=1000(alekhya) gid=1000(alekhya) groups=1000(alekhya),27(sudo),100(users)  
alekhya@Alekhyareddy:~$ ls  
LICENSE      Week2      fork_exec.c  main.c      src  
Makefile     assets    fork_program  obj         tests  
OSLab       bin       hello        process       
OSSP       docs      hello.c      process.c  
README.md   fork_exec include      scripts  
alekhya@Alekhyareddy:~$ history  
1 id alekhya  
2 wsl --version  
3 where ubuntu  
4 whoami  
5 nano ~/.bashrc  
6 echo $$  
7 pstree  
8 ps -ef  
9 sleep 60 &  
10 ps -ef | grep sleep  
11 strace ls  
12 /bin/ls  
13 exec ls  
14 nano src/fork_exec.c  
15 ls  
16 cd OSSP  
17 ls  
18 gcc fork_exec.c -o fork_exec  
19 cd OSSP  
20 tree  
21 ls  
22 nano fork_exec.c  
23 gcc fork_exec.c -o fork_exec  
24 ./fork_exec  
25 mkdir OSSP  
26 cd OSSP  
27 mkdir OSSP  
28 git init
```

```
alekhya@Alekhyareddy: ~  ×  +  ∨  
68 mkdir scripts  
69 mkdir assets  
70 cd ~/OSSP  
71 git init  
72 mkdir src  
73 ls src  
74 nano src/fork_exec.c  
75 mkdir OSSP  
76 cd OSSP  
77 git init  
78 mkdir src include obj bin docs tests scripts assets  
79 touch .gitignore README.md LICENSE Makefile  
80 touch src/main.c src/shell.c src/parser.c src/executor.c src/builtin.c  
81 touch include/shell.h include/parser.h include/executor.h include/builtin.h  
82 touch docs/design.md docs/report.pdf  
83 touch tests/test_parser.c tests/test_executor.c  
84 touch scripts/run.sh  
85 touch assets/architecture.png  
86 tree  
87 ls OSSP  
88 rmdir OSSP  
89 ls -la OSSP  
90 mkdir OSSP  
91 cd OSSP  
92 git init  
93 mkdir src include obj bin docs tests scripts assets  
94 touch .gitignore README.md LICENSE Makefile  
95 touch src/main.c src/shell.c src/parser.c src/executor.c src/builtin.c  
96 touch include/shell.h include/parser.h include/executor.h include/builtin.h  
97 touch docs/design.md docs/report.pdf  
98 touch tests/test_parser.c tests/test_executor.c  
99 touch scripts/run.sh  
100 touch assets/architecture.png  
101 tree  
102 pstree  
103 ps -ef  
104 echo $$  
105 strace ls  
106 nano src/fork_exec.c  
107 gcc /fork_exec.c -o /fork_exec  
108 gcc fork_exec.c -o fork_exec  
109 wsl --status  
110 pwd  
111 date  
112 whoami  
113 hostname  
114 uname  
115 id  
116 ls  
117 history  
alekhya@Alekhyareddy:~$ |
```